SUJITH P

VLSI Physical Design Engineer | Analog & Mixed-Signal IC Implementation

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PROFESSIONAL SUMMARY

Pre-final year ECE undergraduate specializing in CMOS device physics, analog circuit design, and VLSI physical design. Hands-on experience with parasitic-aware layout implementation and DRC/LVS verification using Cadence EDA tools. Completed full RTL-to-GDSII flow with comprehensive PPA analysis. Presented research at ISRO-affiliated international conference. Seeking full-time/internship roles in Analog Physical Design, Mixed-Signal IC Implementation, or Digital VLSI Design at India's leading semiconductor and fabless companies.

TECHNICAL SKILLS

HDL / Languages	Verilog, SystemVerilog (Design & Verification), SystemC, C, Python
Cadence EDA Tools	Xcelium (Simulation) • Genus (Synthesis) • Innovus (P&R) • Quantus (Parasitic Extraction) • Pegasus (DRC/LVS) • Virtuoso (Analog Layout)
Other EDA Tools	OpenROAD, Xilinx Vivado
Core VLSI Concepts	CMOS Device Physics • Analog Circuit Design • Differential Pair & Current Mirror • Common Centroid & Interdigitation • Parasitic & RC Delay Analysis • DRC/LVS Verification • Layout Dependent Effects (LDE) • Static Timing Analysis • RTL-to-GDSII Flow • PPA Optimization
Dev & Productivity	Git / GitHub, VS Code, MS Word, MS Excel

INTERNSHIP EXPERIENCE

Digital Electronics & VLSI Intern | Codec Technologies Pvt. Ltd., Coimbatore *Nov 2025 – Feb 2026*

- Executed RTL design, functional simulation, and synthesis using Cadence Xcelium and Genus for industry-level digital modules.
- Performed backend VLSI tasks: placement, clock tree synthesis, routing, and PPA trade-off analysis using Cadence Innovus.
- Verified timing closure through static timing analysis; improved critical path delay by design restructuring.
- Strengthened expertise in CMOS design principles and end-to-end VLSI implementation methodology.

KEY PROJECTS

Low-Power Vedic Multiplier – Full RTL-to-GDSII Implementation *[Individual]*

- Designed a low-power Vedic multiplier in Verilog and executed the complete RTL-to-GDSII flow using Cadence Genus (synthesis) and Innovus (place & route).
- Achieved measurable improvement in power efficiency and critical path delay versus conventional multiplier architectures through switching activity optimization.
- Ran parasitic extraction with Cadence Quantus and iterated the layout to minimize RC delay; verified design with DRC/LVS using Pegasus.

VLSI-Enabled AI Automation for Fault Detection in Aerospace Systems *[Team – 3 Members | Conference Paper]*

- Architected a VLSI-based intelligent monitoring system for real-time fault detection, integrating AI-driven predictive algorithms with hardware control logic.
- Designed and simulated RTL modules targeting aerospace-grade robustness; analyzed PPA trade-offs for high-availability design.

- Research published and presented at the International Conference on Advances in Aerospace and Energy Systems (IAES-2026), ISRO Propulsion Complex, Mahendragiri.

Low-Power Real-Time Speech Recognition System *[Team – 3 Members]*

- Designed a hardware-efficient speech recognition architecture for low-power, real-time edge processing.
- Integrated signal preprocessing and feature extraction modules with embedded control logic; analyzed latency and power trade-offs for energy-efficient operation.

ACHIEVEMENTS & RECOGNITION

- **Conference Presenter:** Presented VLSI & AI research paper at IAES-2026 held at ISRO Propulsion Complex, Mahendragiri; certified by Indian Society for Heat and Mass Transfer (ISHMT-RCT).
- **NIT Jamshedpur Hackathon:** Secured Top 30 Rank in AI/ML-Powered FPGA Design & Simulation Hackathon at Industry Academia Conclave 2025, NIT Jamshedpur, in collaboration with VLSI For All.
- **Resource Person:** Invited speaker at Mahendra Engineering College (March 2026) on 'High Performance System Design using EDA Tools' — delivered Cadence-based VLSI design workflows to B.E. students.
- **1st Prize – INNOVAX 2025:** Won first place at Mini Project Expo for 'Lung Cancer Detection' project, organized by Dept. of ECE, Mahendra Engineering College.

CERTIFICATIONS & TRAINING

- **Cadence RTL-to-GDSII Flow v6.0** — 2026
- **SystemVerilog for Design and Verification** — 2025
- **SystemC Language Fundamentals v12.2** — 2026
- **VLSI for Beginners – 7-Day Training Program** — VLSI For All, Jan 2026
- **Semiconductor Circuit Design and Simulation Training** — VIT Vellore, sponsored by British Council, Jan 2026
- **VLSI Frontiers: Exploring Cadence Tools Workshop** — ECE Association, PSG Institute of Technology and Applied Research, Coimbatore, 2025

EDUCATION

B.E. Electronics and Communication Engineering *2022 – 2026*
Mahendra Engineering College, Namakkal | CGPA: 7.63 (up to 6th Semester)

Class XII – Higher Secondary *2022*
SBV Matric Higher Secondary School, Thuraiyur | 65%

Class X – Secondary School *2020*
Sowdambikaa Matric Higher Secondary School, T.pet